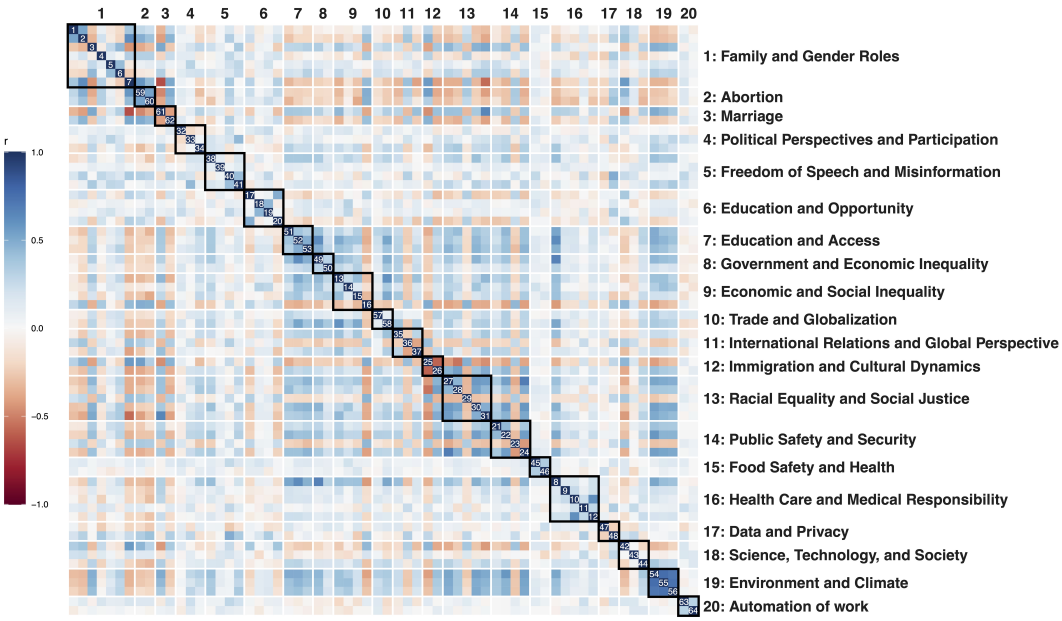

Figure 1: Correlations among self-reported human attitudes. Attitudes (labeled along the diagonal) are ordered by topic area (labeled on axes), and topic areas are separated by grid lines. Black bounding boxes highlight topic groupings along the diagonal indicating within-topic associations.

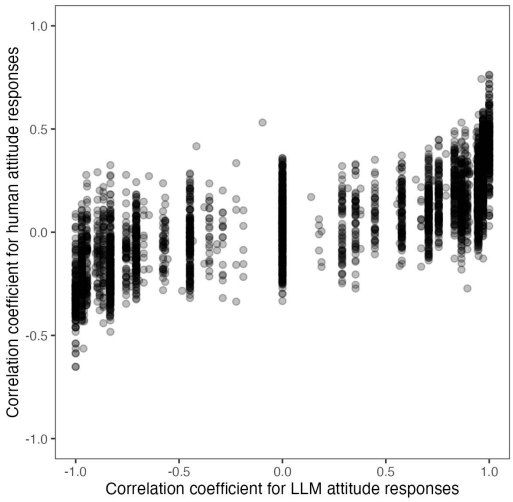

Figure 2: Scatterplot of correlation values observed in human data against correlation values among GPT-4o estimations.

peaks near -1 and 1. Figure 2 illustrates this pattern in the clustering of points at the extremes along the x-axis.

Semantic Similarity and Estimated Correlations Next, we examined the extent to which GPT-4o's predicted associations between attitudes rely on the semantic similarity between those attitudes. Cosine similarity of the statements was positively correlated with the absolute strength of association among statements ($r = .33$, 95% CI $[.30, .75]$, $p < .001$). This relationship is quite imperfect, so that dissimilar statements sometimes have strong associations, and similar statements sometimes have only weak associations.

As GPT-4o reliably overestimates the strength of associations, in the following analyses we rank-transform both human and LLM-predicted correlations for comparison purposes. Figure 3 shows the relationship between the rank-differences comparing human and LLM-estimated attitude correlation against their statement's cosine similarities. A modest negative trend is observed, indicating that model-estimated correlations are more reflective of human correlations among more-similar item pairs.

A tendency to leverage similarity appears to bias GPT-4o's correlation estimates more generally: Regressing the absolute human correlation values on cosine similarity for each pair, we divide attitude-pairs into those more and less-strongly associated than would be predicted by cosine similarity. We find that GPT-4o more-commonly overestimates the association for pairs that are less-strongly associated than predicted from cosine similarity (64.9% llm-based-rank $>$ human-

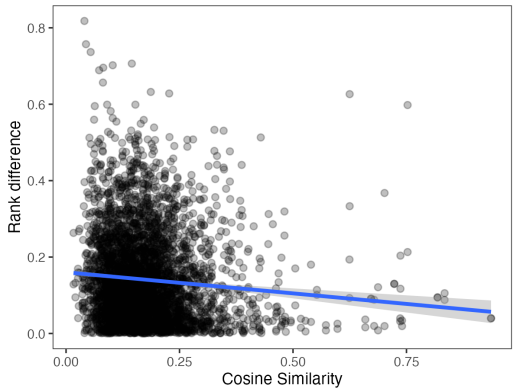

Figure 3: Scatterplot comparing differences between rank-transformed correlation values observed in human data and estimated by GPT-4o against the cosine distance of each statement pair. A regression line highlights the negative linear relationship.

rank) and underestimates associations for pairs that are more-strongly associated (70.5% llm-based-rank $<$ human-rank).

Nevertheless, the biases induced by semantic similarity appear to have only modest impacts on GPT-4o's ability to recapitulate correlations among attitudes. Filtering for only dissimilar (i.e. $S_c < .20$) attitude pairs significantly reduces the correlation between GPT-4o-estimated correlations and human correlations ($p = .001$), yet the relationship between estimated and observed correlations remained strong ($r = .724$).

Study 3: Predicting human responses using GPT-4o

Finally, we examined GPT-4o's ability to predict individual respondents' answers to each target attitude from their responses to other attitude questions. Crucially, we are interested in examining the degree to which GPT-4o might rely upon the semantic similarity of the target attitude and the attitudes in its prompt.

Selecting and evaluating top-k predictors

To test this, we examined GPT-4o's ability to predict target attitudes using two feature selection methods. First, following Hwang et al. (2023), we identified a set of *semantically-similar* predictors for each target attitude by selecting the top-k most similar attitudes based on cosine similarity (with $k \in \{3, 8\}$). For each target, we also identified a set of *semantically-dissimilar* predictors. Here, we exclude any predictors with $S_c > .20$ and from the remainder choose the top-k most correlated attitudes based on human data.

"Oracle" model training

For each target attitude and set of semantically-similar and dissimilar predictors, we trained a random forest model to examine how well those predictors could in-fact predict the target item. We evaluate the model fit using a 10-fold cross validation procedure, evaluating predictive accuracy using the model predictions on the held-out validation splits. We take these models as "oracle" models and we take their validation performance to represent the upper-bound of predictability for the target attitudes from the selected predictors.[3]

Generating GPT-4o predictions

To predict each participant's responses, we constructed prompts containing 1) instructions for the model to "Help predict a person's answers on a social attitudes survey"; 2) a set of k (i.e., 3 or 8) example statements and a participant's actual responses observed in Study 1; 3) instructions to generate a prediction based on those statement-answer pairs; and 4) the target statement along with the list of answer choices (i.e., the five agreement levels). For each item, we composed predictor sets for the top-3 and top-8 items chosen using our criteria for selecting semantically-similar and semantically-dissimilar items. We then passed the prompts to GPT-4o through the OpenAI API.

Metrics

Accuracy For GPT-4o, we define accuracy as the exact match between the model's output and the choice text.

Results

Table 1 shows the predictive accuracy of "oracle" models and GPT-4o based on the top-k semantically-similar and semantically-dissimilar item selection strategies. As shown by the oracle model results, the semantically-similar items were in-fact more predictive of the target attitudes. This is to be expected: semantically-similar items often have relatively strong correlations with the target, so that excluding these items puts the dissimilar models at a substantial disadvantage, despite choosing the dissimilar items based on observed correlations. When ncluding all items, GPT-4o's predictions were more accurate when prompted with similar items.

Although the semantically-similar items were more predictive overall, for a substantial portion of target attitudes ($24/64$ for $k = 3$ and $14/64$ for $k = 8$), oracle models using semantically-dissimilar items were superior to those using similar items. Again this is generally to be expected: the most semantically-similar items are not *always* those that are most predictive. Importantly, these cases offer strong a test of the role of semantic similarity in informing language models' attitude predictions: If the language model predicts as well or better with low-similarity items, this would then demonstrate that it is not relying on simple semantic similarity. However, if its predictions are worse in these cases where they could

[3] We qualify this label with a potential limitation: given that these models are estimated from a limited sample of human data, superior predictive accuracy may technically be possible.

Table 1: Predictive accuracies of GPT-4o and "Oracle" Random Forest models.

Model	Target subset	Selection method	Top-k	Accuracy
GPT4o	all	similar (S_c)	3	43.5%
			8	45.2%
		dissimilar (r)	3	40.4%
			8	41.9%
GPT4o	dissimilar > similar	similar (S_c)	3	41.2%
			8	40.8%
		dissimilar (r)	3	42.0%
			8	42.2%
Oracle	all	similar (S_c)	3	47.7%
			8	49.6%
		dissimilar (r)	3	44.4%
			8	45.5%
Oracle	dissimilar > similar	similar (S_c)	3	43.9%
			8	45.1%
		dissimilar (r)	3	47.3%
			8	47.6%
Chance	all	–	–	29.4%

Note: Models are evaluated against all observed responses from all participants (64 items $\times$ 376 participants $= 24,064$).

or should be better, then we can conclude it is in some way relying on semantic similarity.

Focusing on just these items, the results in Table 1 demonstrate that GPT-4o is also more accurate when predicting from the dissimilar items than similar items. We take this as a clear demonstration that GPT-4o is capable of making meaningful social inferences in the absence of semantic similarity between statements. At the same time, comparing the accuracy of these two predictor sets between GPT-4o and the oracle models fit to human data reveals that GPT-4o's predictions are closer to the upper-bound of predictability when it is prompted with semantically-similar items (by 2.6% for $k = 3$ and 1.1% for $k = 8$). This suggests that GPT-4o does, at the same time, leverage the more surface-level similarity between attitudes in generating predictions.

Discussion

Using a novel dataset of human attitude across a range of social topics, we tested the ability of frontier language models (i.e. GPT-4o) to estimate the correlations among human attitudes and to predict individuals' attitudes from one another. Our findings demonstrate that GPT-4o can largely replicate the inter-correlations observed in human data, and can meaningfully predict individuals' attitudes. Going beyond prior findings (Hwang et al., 2023), we show that LLMs attitude predictions are not merely driven by surface-level semantic similarity, but instead tap in to latent associations among human attitudes.

Despite these novel findings, our study has a limitation that we only calculated model estimation alignment and prediction accuracy based on five levels of outputs for each target-example statement pair. Although we experimented with repeated sampling and prompting for probability distribution outputs, these approaches did not result in much variability or improvement in model estimations.

Across our analyses with GPT-4o, we constrained the model to immediately respond with a predicted response option. This may underestimate the potential performance of the model, and future research could explore other prompting strategies. For instance, model prompts could utilize the chain-of-thought methods (CoT) (Wei et al., 2023) to encourage step-by-step "thinking" from the model, which might improve performance (Long et al., 2024). Alternately, future work might examine the capabilities of new reasoning models such as GPT-o1 (OpenAI, 2024) and DeepThink R1 (DeepSeek-AI et al., 2025). These approaches might improve performance or decrease the biases we observed with respect to semantic similarity. Further, examination of the intermediate CoT or reasoning steps might support deeper insights into the information leveraged by the model. For instance, we might examine whether models perform better when they explicitly infer (and tokenize in their reasoning) the ideology of a target person, or when they take other latent dimensions of attitudes into account.

Our findings underscore a number of existing concerns about the safe deployment of LLMs and other AI systems. First, LLM alignment efforts seeking to personalize models to users' existing attitudes risk creating echo chambers in human-AI interactions, with the potential to increase polarization (Axelrod et al., 2021). In addition, the major objective of monetizing LLM chatbots has driven large tech companies to adopt models for advertisement creation and promotion, e.g. Microsoft's Copilot (Microsoft, 2024). When AI ads are difficult to notice even with ad disclosures, users reported feeling manipulated, intruded, and less trusting towards LLM chatbots (Tang, Sun, Curran, Schaub, & Shin, 2024a).

Moreover, AI safety advocates and researchers have identified a number of capabilities that, if achieved by AI systems, could present substantial risk to humans. Among these are the ability to persuade or manipulate humans (Ji et al., 2024; Burtell & Woodside, 2023). Persuasion often benefits from understanding of others' existing viewpoints (Lewandowsky, Ecker, Seifert, Schwarz, & Cook, 2012; Powell et al., 2023), suggesting that the ability to predict attitudes may be a key aspect of this larger capacity. Our findings therefore underscore existing concerns about the potential risks of AI systems for persuasion and manipulation. Although there may be positive uses of such capabilities (Karinshak, Liu, Park, & Hancock, 2023), these capabilities nevertheless present alarming risks for automated propaganda and targeted misinformation.

Conclusion

Using a novel dataset of human responses to diverse attitude measures, we demonstrated the ability of frontier language models to replicate the interrelations among human beliefs without relying on surface semantic similarities. These results showcase these models' higher-level social reasoning capacity that extends beyond mere pattern matching.